NO FORMAL DATE FOR EXAM. CAN DO QUESTIONNAIRE WHEN I WANT (MIN. 2 PPL)

INTRODUCTION

GOAL OF THE COURSE: NON-INVASIVELY RETRIEVE INFORMATION ABOUT ONE OR MORE OBJECTS IN AN INACCESSIBLE DOMAIN THROUGH EXTERNAL OBSERVATION: DIAGNOSE PRESENCE OF SOMETHING OF INTEREST (APPLICATION DEPENDENT) AND/OR CREATE IMAGE OF INFORMATIONAL CONTENT

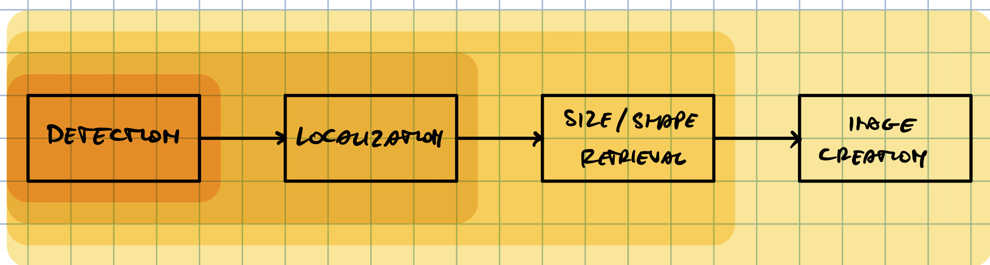
DIAGNOSIS → DIA: THROUGH, GNOSIS: KNOWING

IMAGE: REPRESENTATION OF A PHYSICAL QUALITY FROM THE POV OF THE SENSOR USED, WHERE EACH COORDINATE (PIXEL/VOXEL) HAS A NUMBER/COLOR.

THERE ARE 4 TYPES OF OBJECTIVES:

- DETECTION: IS IT THERE?
- LOCALIZATION: WHERE IS IT?
- SIZE/SHAPE RETRIEVAL: HOW LARGE IS IT? WHAT IS ITS SHAPE?
- IMAGE CREATION: HOW IS IT MADE?

EVERY STEP REQURES INFORMATION FROM THE PREVIOUS STEPS ⇒ AMOUNT OF INFORMATION REQUIRED INCREASES FOR EACH STEP



WAVES

IN GENERAL, WAVES CAN BE USED TO COMMUNICATE INFORMATION AND SENSE A SCENARIO. OUR INTEREST IS IN THE LATTER.

WE ARE INTERESTED IN EM WAVES, BUT THIS APPLIES TO ALL WAVE TYPES (ACOUSTIC, MECHANICAL, EM, ...)

WAVES TO COMMUNICATE

WE CAN TRANSMIT INFORMATION BECAUSE WAVES CARRY ENERGY WHICH IS ASSOCIATED WITH INFORMATION AND THAT ENERGY OF THE WAVE TRAVELS AT THE SAME SPEED OF THE INFORMATION ASSOCIATED TO THE WAVE (NOT OBVIOUS/SELF-EVIDENT).

WITH NO ENERGY, THERE IS NO INFORMATION ← ENTROPY

WAVES TO SENSE

AS WAVES PROPAGATE, THEY CHANGE THEIR BEHAVIOR DEPENDING ON THE MEDIUM THEY PROPAGATE THROUGH. IN PARTICULAR, WAVE VELOCITY DEPENDS ON THE MEDIUM, ALLOWING US TO SENSE DISCONTINUITIES.

ACOUSTIC

$$v = \sqrt{\frac{C}{\rho}}$$

STIFFNESS (C), DENSITY (ρ)

ELECTROMAGNETIC

$$v = \frac{1}{\sqrt{\varepsilon \mu}}$$

DIELECTRIC PERMITTIVITY (ε), MAGNETIC PERMEABILITY (μ)

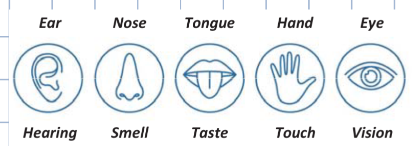
SENSING

SENSING THEREFORE ALLOWS US TO SAY SOMETHING ABOUT THE MEDIUM.

SENSING IMPLIES THE PRESENCE OF SOMETHING CAPABLE OF "READING" THE WAVES: THE SENSOR.
A SENSOR CAN'T "READ" ALL TYPES OF WAVES, IT IS SPECIFIC TO WAVES WITH SPECIFIC CHARACTERISTICS.

FROM THE POV OF THE SENSING SYSTEM, IF THERE IS NO SENSOR, NOTHING EXISTS.

IN THE HUMAN CASE WE HAVE 5 SENSING SYSTEMS:

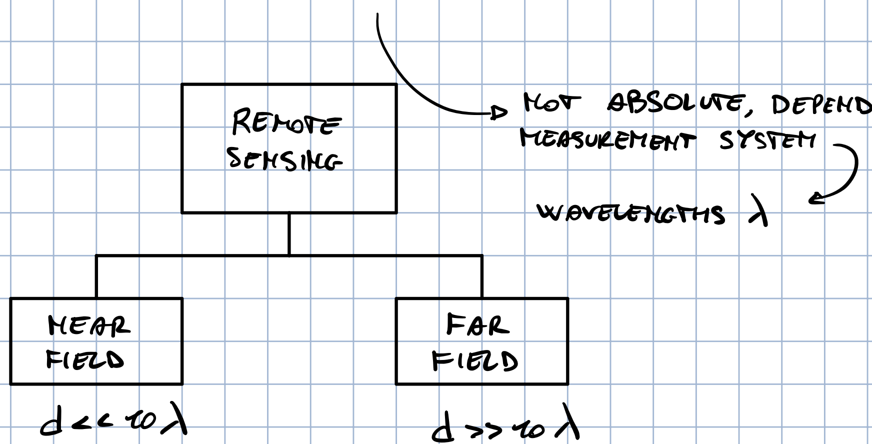


REMOTE SENSING

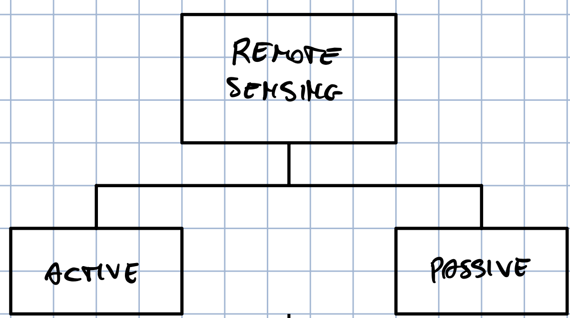
REMOTE SENSING REFERS TO WHEN THE DETECTION IS NOT PERFORMED THROUGH HUMAN SENSES, BUT WITH THEIR REMOTIZED VERSION: SOMETHING THAT EXTENDS HUMAN SENSES.

E.G.: TOMOGRAPHY IS NOT PERFORMED THROUGH HUMAN VISION, BUT THE RESULTING IMAGE IS SEEN THROUGH OUR EYES

REMOTE SENSING CAN BE SUBDIVIDED BASED ON THE DISTANCE OF THE DETECTOR WITH RESPECT TO THE INVESTIGATION DOMAIN

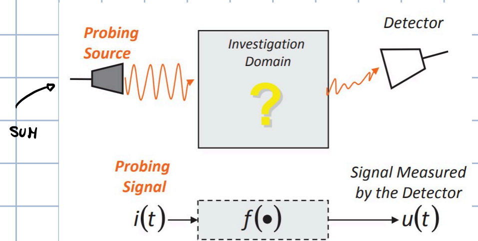


REMOTE SENSING CAN BE FURTHER SUBDIVIDED BASED ON THE PRESENCE OF A PROBING SOURCE.

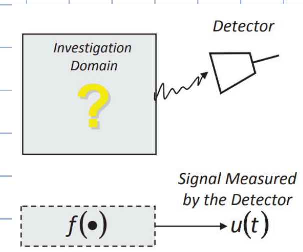


THE DETECTOR COLLECTS THE SIGNAL GENERATED BY THE INTERACTION BETWEEN THE SOURCE AND THE INVESTIGATION DOMAIN.
NOTE THAT THE SOURCE DOESN'T NECESSARILY NEED TO BE PART OF THE MEASUREMENT SYSTEM

COULD BE SUN



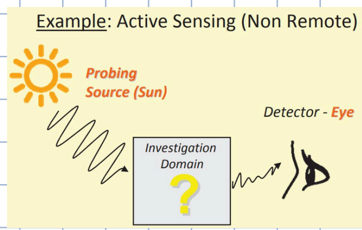
THERE IS NO PROBING SOURCE, THE DETECTOR RELIES ON THE SIGNAL DIRECTLY EMITTED BY THE SCENARIO IN THE INVESTIGATION DOMAIN



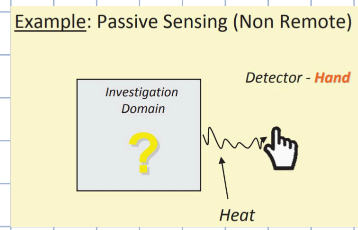
E.G. BLACK BODY RADIATION SENSING

$f(i(t), \text{SCENARIO})$

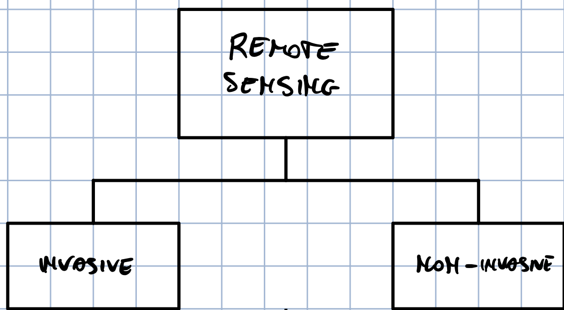
DEPENDS ON INPUT AND SCENARIO



$f(\text{SCENARIO})$



A FURTHER CLASSIFICATION IS DONE BASED ON THE POSITION OF THE DETECTOR WITH RESPECT TO THE INVESTIGATION DOMAIN



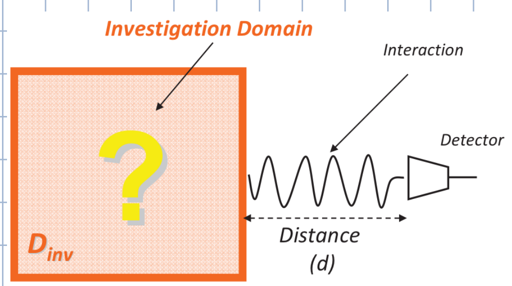
THE DETECTOR IS INSIDE THE INVESTIGATION DOMAIN

THE DETECTOR IS OUTSIDE THE INVESTIGATION DOMAIN

DOMAINS AND ACTORS

INVESTIGATION DOMAIN

THE INVESTIGATION DOMAIN IS THE REGION OF SPACE THAT WE ARE ANALYZING AND WHERE WE SUPPOSE THE OBJECT OF ANALYSIS IS.

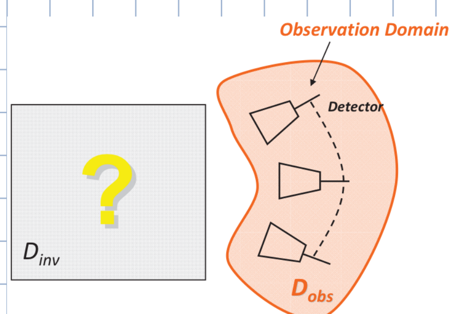


WE REFER TO ANYTHING THAT IS CONTAINED INSIDE THE INVESTIGATION DOMAIN AS "SCENARIO"

OBSERVATION DOMAIN

THE OBSERVATION DOMAIN IS THE REGION OF SPACE FROM WHICH WE ARE LOOKING AT THE INVESTIGATION DOMAIN. IT IS THE REGION IN WHICH THE DETECTOR/SENSORS, THE DEVICE THAT COLLECTS THE SIGNAL, IS PLACED.

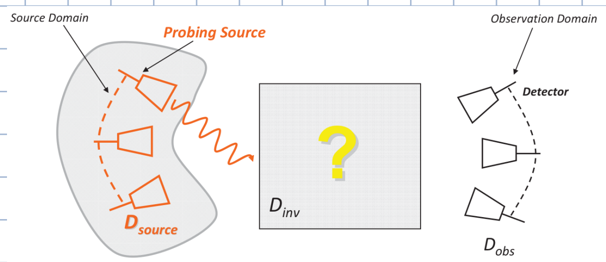
THE DETECTOR IS WHAT COLLECTS THE SIGNAL. IT MUST BE SUITABLE FOR THE SELECTED APPLICATION.



THIS REGION CAN EITHER BE OUTSIDE OR INSIDE THE INVESTIGATION DOMAIN, DEPENDING ON WHETHER THE SENSING IS INVASIVE OR NON-INVASIVE.

SOURCE DOMAIN

THE SOURCE DOMAIN DESCRIBES THE SPATIAL REGION THAT DEFINES THE PROBING SOURCES: THE DEVICES THAT GENERATE THE PROBING SIGNAL.



AMIRS PROBLEM

ACTIVE NON-INVASIVE REMOTE SENSING

NON-INVASELY RETRIEVE INFORMATION ABOUT SOMETHING WHICH IS INSIDE AN INACCESSIBLE DOMAIN THROUGH EXTERNAL MEASUREMENTS.

MEANING OF EXISTING

IT IS COMMON SENSE THAT AN OBJECT EXISTS INDEPENDENTLY FROM SOURCE, DETECTOR AND INTERACTION, HOWEVER THIS IS NOT TRUE IN REMOTE SENSING.

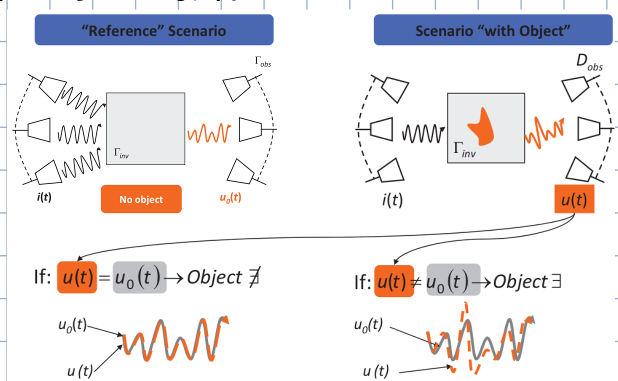
FROM THE POV OF AN AMIRS SYSTEM, AN OBJECT EXISTS ONLY IF:

- THE OBJECT IS PRESENT (AS PER COMMON SENSE)
- THERE IS A PROBING SOURCE
- THE OBJECT INTERACTS WITH THE PROBING SIGNAL
- THE INTERACTION IS DIFFERENT FROM THAT WITHOUT THE OBJECT ⇒ DIFFERENTIAL ANALYSIS
- THERE IS A DETECTOR CAPABLE OF DETECTING THE INTERACTION

DIFFERENTIALITY

WE DON'T LOOK AT ABSOLUTE VALUES TO DETECT OBJECTS, WE LOOK AT HOW DIFFERENT THE INTERACTION IS FROM A REFERENCE SCENARIO (I.E. ONE IN WHICH WE KNOW THE OBJECT WE ARE LOOKING FOR IS NOT PRESENT).

THE REFERENCE SIGNAL REPRESENTS THEREFORE THE INTERACTION BETWEEN THE SOURCE AND THIS REFERENCE SCENARIO.



RECOGNITION

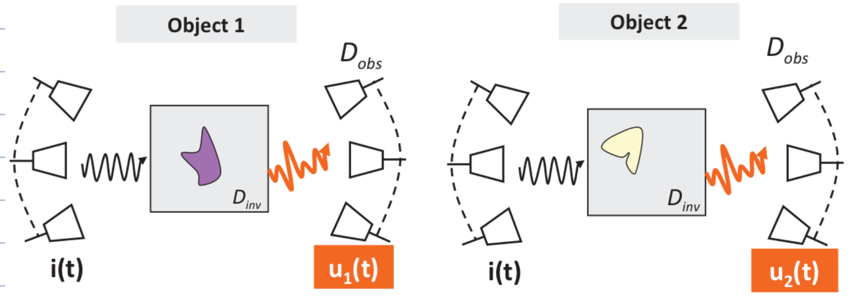
SAYING THAT SOME OBJECT EXISTS IS DIFFERENT FROM BEING ABLE TO DISCERN MULTIPLE DIFFERENT OBJECTS (⟶ RECOGNITION).

TO DISTINGUISH OBJECTS WE NEED TO VERIFY THEIR UNIQUENESS.

FROM THE POV OF AN AMIRS SYSTEM, TWO OBJECTS ARE DIFFERENT ONLY IF:

- THE INTERACTIONS WITH THE PROBING SIGNAL ARE DIFFERENT

$$OBJ_1 \neq OBJ_2 \iff u_1 \neq u_2$$



Object 1
D_{obs}
D_{inv}
i(t)
$u_1(t)$
Object 2
D_{obs}
D_{inv}
i(t)
$u_2(t)$